Huy Hoang

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Education

University of Florida **Expected Graduation**: May 2027
Bachelor of Science in Computer Engineering | GPA: 3.9/4.0 *Gainesville, FL*

Technical Skills and Certifications

Languages: Python, Go, C++, Java, Shell, SQL, Protobuf, TypeScript, JavaScript
Frameworks: FastAPI, Flask, Spring, Node.js, React Native, React
Developer Tools: AWS, Terraform, Kubernetes, Kafka, MySQL, Docker, Jenkins, OpenSearch, Cadence
AI/ML: Model Context Protocol (MCP), LangChain, PyTorch, TensorFlow
Libraries: OpenCV, NumPy, pandas, COLMAP
Certifications: NVIDIA Fundamentals of Deep Learning

Experience

Software Engineering Intern May 2026 – July 2026
Uber

- Led search improvements for an A/B experimentation platform end-to-end (design doc, bug bash, production readiness review, rollout), engineered OpenSearch-native sort and filter on Go gRPC endpoints for 120k+ experiments with 27k+ searches/day
- Ran a zero-downtime schema migration in Go across protobuf, MySQL, Kafka, and OpenSearch indices, dual-writing on the live path and backfilling 120k+ production experiments through idempotent Cadence workflows at 99.8% completion
- Built a Go service exposing 7 experiment-management tools over Model Context Protocol (MCP), integrating it into an agentic debugging pipeline that root-causes misconfigured experiments for 1k+ services running experiments
- Architected a Python LLM reviewer for A/B experiment configs across 500 diffs/day, encoding past-incident patterns as lazy-loaded rule packs that cut agent context 50%, and built the eval harness measuring 94% precision and 86% recall

Machine Learning Research Assistant Sep. 2024 – Sep. 2025
University of Florida – AI Lab

- Developed GPU-accelerated PyTorch pipelines for dense feature matching, enabling high-throughput 3D inference
- Architected a distributed pipeline consolidating multi-view 3D reconstructions into a unified field-scale model
- Applied monocular depth estimation to densify point clouds, achieving 90%+ data completeness for volumetric analysis

Software Engineer Intern May 2025 – Aug. 2025
FPT Corporation

- Implemented RESTful APIs in Java Spring Boot with WSO2 mediation policies, deployed with AWS EC2, Docker, and monitored via Grafana and Prometheus, achieving 98% request accuracy and improving response time by 20ms
- Configured and integrated 12+ enterprise microservices using WSO2 API Gateway deployed on Kubernetes, reducing cross-service failures by 20% and increasing system resilience in production clusters
- Developed custom Java WSO2 handlers for payload encryption/decryption and token validation using JWT and OAuth2, improving gateway throughput by 25% and enforcing secure, authenticated API communication across teams

Site Reliability Engineer Intern May 2024 – Aug. 2024
FPT Information System

- Automated AWS EKS infrastructure provisioning with Terraform and GitOps, improving deployment efficiency by 30%
- Built CI/CD pipelines with Jenkins and Docker, enabling instant rollbacks and cutting deployment time by 90%
- Deployed Prometheus monitoring across 20+ EC2 instances, cutting network setup from 2h to 15min and crash rates by 30%

Teaching Assistant Aug. 2024 – Present
University of Florida (Computer Organization & Fundamentals Programming 2)

- Led mini-lectures and structured office hours for 50+ students; section grades averaged 15% higher than other section

Projects

Horus' Eyes AI Platform (1st Place, SwampHacks) | *Terraform, AWS, YOLOv8, Flask, Docker*

- Awarded 1st Place for Best Use of Terraform by automating cloud infrastructure for high-throughput ML model deployment
- Engineered an inference pipeline leveraging YOLOv8 and DPT for object and text detection to assist visually impaired users
- Deployed a low-latency Flask + AWS Bedrock pipeline on EC2, executing real-time Text-to-Speech generation

Context-Aware LLM Application | *LangChain, AWS Bedrock, Python, FastAPI, Docker*

- Developed a generative AI application utilizing LangChain and AWS Bedrock with optimized state and memory management
- Implemented lightweight context caching to ensure highly accurate, multi-turn conversations without latency degradation

Leaderships

Software and Electrical Team Lead | *Society of Asian Scientists and Engineers (SASE) Engineering Team*